

Memo

Date: Tuesday, December 20, 2022

Project: B2280 – Chilled Water System WWYK200131

HDR Proj. # 10296383 Task 006.24.1

To: Project Team

From: Thomas C. Wesling P.E.

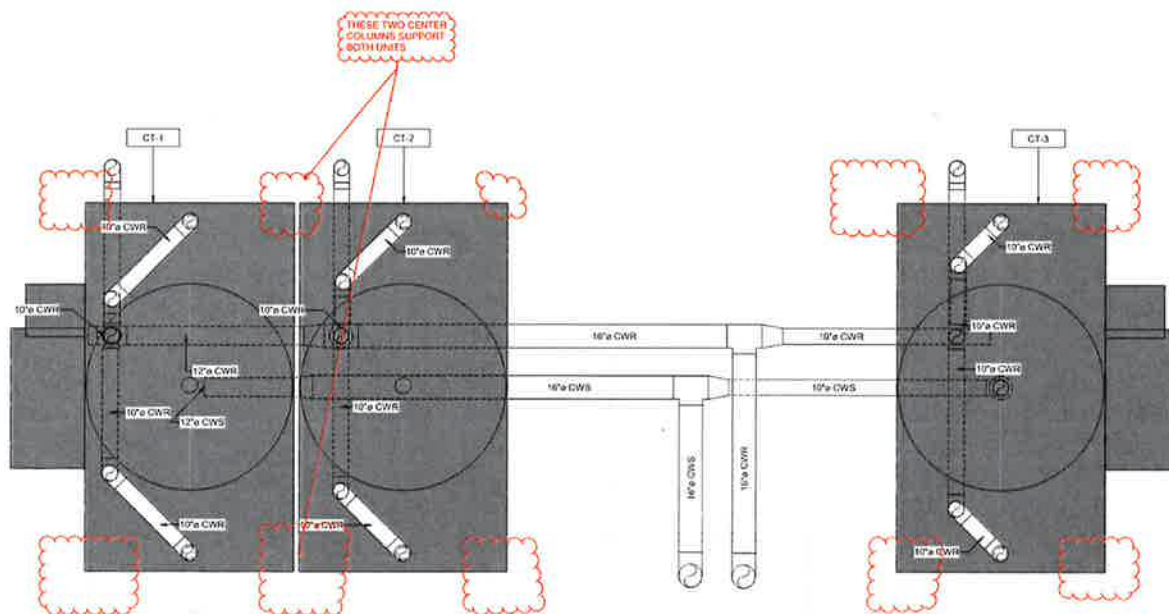
Subject: Drilled Shaft Foundation Recommendations



Thomas C. Wesling

Project Description

The proposed project is located within Tinker AFB in Oklahoma City, Oklahoma. The proposed new chillers will be located on the east side of Building B2280 in the general location of the existing chillers which will be removed. Building 2280 is located northwest of the intersection of South Douglas Boulevard and Southeast 59th Street. Based on discussions with the HDR project structural engineer, the project includes three new chillers that will be column supported on drilled shafts as shown below. The site is located between two existing buildings, so site grade is expected to remain near the existing conditions.



Summary of Field Investigation and Laboratory Testing Program

Standard Testing and Engineering Company, LLC (STEC) recently completed a geotechnical data study for the proposed chillers. STEC's geotechnical data report for the project (STEC Project No. 2230-0534) is dated October 27, 2022 and is attached to this memo. The geotechnical data study included two 51-ft deep borings (B-1 and B-2). The laboratory testing

program included water contents, Atterberg limits, grain size analyses, unconfined compression test, and analytical tests. The subsurface exploration report prepared by STEC is attached to this design memo includes a project vicinity map, boring location plan, boring logs, and laboratory test results.

Subsurface and Depth-to-Water Conditions

Subsurface conditions can best be understood by a thorough review of the boring logs in the attached subsurface exploration report. At the surface 4 inches of asphaltic concrete pavement was encountered over 6 inches of aggregate base at each of the two borings. Below the pavement materials, the two 51-ft deep borings generally encountered 5 strata that are summarized below.

Stratum 1. Reddish brown lean clay with various amounts of sand and asphaltic concrete debris was encountered below the aggregate base and extends to a depth of 8 feet at boring B-1. Reddish brown sandy lean clay over reddish brown silty sand with asphaltic concrete debris was encountered below the aggregate base and extends to a depth of 11 feet at boring B-2. Stratum 1 has measured liquid limits varying from 16 to 38, plasticity indices ranging from non-plastic to 24, percent passing the No. 200 sieve varying from 24 to 92, and SPT N-values ranging from 15 to 25 blows per foot (bpf).

Stratum 2. Reddish brown weathered shale was encountered below the upper soils. The weathered shale extends to depths of 17 and 16 feet at the boring B-1 and B-2 locations, respectively. Stratum 2 has measured liquid limits of 26 and 34, plasticity indices of 15 and 19, percent passing the No. 200 sieve of 53 and 96, and SPT N-values of 34, 37, and 47 bpf.

Stratum 3. Reddish brown shale was encountered below the weathered shale. The shale extends to depths of 25 and 31 feet at the boring B-1 and B-2 locations, respectively. Stratum 3 has a measured liquid limit of 32, plasticity index of 19, percent passing the No. 200 sieve of 85, and SPT N-values ranging from 95 blows for 8 inches of penetration to 50 blows for 3 inches of penetration.

Stratum 4. Reddish brown loosely cemented sandstone was encountered below the shale. The sandstone extends to a depth of 50 feet at each boring location. Stratum 4 has a measured SPT N-values ranging from 50 blows for 2.5 to 5 inches of penetration.

Stratum 5. Reddish brown shale was encountered below the sandstone in the split-spoon sample driven from 50 to 51 feet. Stratum 5 has a measured SPT N-values of 40 blows for the initial 6-inch seating then 50 blows for 4.5 to 5.5 inches of penetration.

The borings were drilled dry with hollow stem augers. Free water was not observed in either boring during or at the completion of drilling operations. However, it should be noted that shallow groundwater is common on Tinker AFB.

Drilled Shaft Design Recommendations

Based on information provided by the project structural engineers the maximum axial compression column load will be on the order of 14 kips. Drilled shafts should be designed in accordance with the flowing recommendations.

1. The structural loads should be carried on straight-sided drilled shafts founded at least 5 feet into the shale in Stratum 3 (22-ft length from ground surface) and sized using the allowable skin friction and end bearing values in the table below.

Drilled Shaft Axial Compressive Design Parameters			
Depth Below Ground Surface * (feet)	Material Type	Allowable Skin Friction (psf)	Allowable End Bearing (psf)
0 – 11	Soil w/ AC Debris	neglect	neglect
11 – 17	Weathered Shale (Clay)	750	neglect
17 – 30	Shale	1,500	12,000
30 – 50	Sandstone	2,500	24,000

2. Tensile (uplift) capacity may be calculated using 70 percent of the allowable skin friction values provided in the table above for each stratum. Buoyant unit weight of the concrete may be added in the tensile capacity.
3. Drilled shafts should have a diameter of 24 inches or greater.
4. The structural capacities of the drilled shafts should be checked for allowable stresses in the concrete, total downward axial loads, tension forces, lateral forces, and moments produced by dead plus probable maximum live loads. Shaft reinforcement should extend from the top to the bottom of the shafts.
5. Maintain a minimum center-to-center spacing between drilled shafts (including existing shafts) of at least three shaft diameters. If the minimum spacing cannot be maintained, the geotechnical engineer should be retained to consider the group effect of closely spaced shafts.
6. Maximum total vertical movement of drilled shafts designed and constructed in accordance with these recommendations is anticipated to be less than 1 inch.

Design Parameters for Lateral Load Analyses

Ultimate soil design parameters are provided herein for use with the computer program LPILE by Ensoft. This computer program is commonly used to evaluate deflections, moments, and shear forces due to lateral loads and moments. The structural engineer should consider head shaft fixity condition and stiffness condition (cracked or uncracked) and select the required length, diameter, and percent of steel for drilled shafts with regard to lateral and axial loads. Recommended input values for the computer program LPILE are provided in the following table.

LPILE Ultimate (Unfactored) Soil Parameters						
Depth Below Ground Surface * (feet)	LPILE Soil / Rock Type	Effective Unit Weight (pcf)	Undrained Cohesion (psf)	Friction Angle (degrees)	Soil Modulus k (pci)	Strain Factor E ₅₀
0 – 5	Soft Clay (Matlock)	120	10	na	na	**
5 – 11 ***	Stiff Clay w/o FW	60	1,200	na	na	**
	Sand (Reese)	60	na	30	**	na
11 – 17	Stiff Clay w/o FW	65	3,000	na	**	**
17 – 30	Stiff Clay w/o FW	65	6,000	na	**	**
30 – 50	Sand (Reese)	70	0	40	**	**
* Depth below ground surface at the boring location.						
** Use LPILE default values.						
*** Each of the two soil types should be evaluated to achieve conservative design.						

Drilled Shaft Construction Recommendations

Drilled shafts should be constructed in accordance with the Unified Facilities Guide Specifications UFGS-31 63 29 Drilled Concrete Piers and Shafts. UFGS-31 63 29 indicates to install drilled shaft foundations in accordance with current versions of ACI 336.1 and FHWA NHI-10-016. Drilled shaft operations should be inspected, on a full-time basis, to (a) verify plan depth and penetration into the bearing stratum, (b) verify shaft dimensions and proper reinforcement, (c) monitor cleanness and amount of water in shaft excavations, (d) monitor placement of concrete and use of tremie or pumps, (e) monitor the extraction of casing, if used, and (f) maintain accurate records.

The existing fill and/or silty sand may not remain stable during drilled shaft construction resulting in the need to use casing and/or slurry for drilled shaft construction. Further, groundwater was not encountered during drilling of the geotechnical boring, however shallow ground water is common at Tinker AFB and should be planned for during drilling shaft construction.

Soil Corrosion Potential

Steel and concrete elements in contact with soil, whether part of a foundation or part of the supported structure, are subject to degradation due to corrosion or chemical attack. Therefore, buried steel and concrete elements should be designed to resist corrosion and degradation based on accepted practices. A series of corrosion tests including pH, sulfate ion concentration, chloride ion concentration, and electrical resistivity were performed on three selected soil samples to evaluate the corrosion potential of the soils at the site. The corrosion potential test results are summarized in the attached geotechnical data report.

Corrosion potential of steel is influenced by electrical resistivity, chloride ion concentration, and pH. Corrosion of steel is more likely in soil environments with low resistivity, high chloride ion concentrations, or low pH. Some general guidelines concerning the corrosion potential of soil on steel as a function of soluble chloride, electrical resistivity, and pH are presented in the following table.

Soluble Chloride Concentrations (ppm)	Electrical Resistivity (ohm-cm)	pH	Corrosion Potential
> 500	0 - 1,000	0 - 4.5	Very Severe
100 - 500	1,000 - 2,000	4.5 - 5.5	Severe
25 - 100	2,000 - 5,000	5.5 - 6.5	Moderate
10 - 25	5,000 - 10,000	> 6.5	Mild
-----	10,000 +	-----	Very Mild

Each variable should be used independently of the others when evaluating soil corrosion potential. For example, it is not necessary to have both an electrical resistivity less than 1,000 ohm-cm and a pH less than 4.5 to indicate a very severe corrosion potential. The laboratory test results for this study indicate the following.

- Measured soluble chloride contents ranged from 11 to 25 ppm, which indicate the soils have a mild corrosion potential.
- Measured electrical resistivity values varied from 1,622 and 4,919 ohm-cm, which indicate the soils have a moderate to severe corrosion potential.
- Measured pH values ranged from 7.1 to 8.9, which indicate the soils have a mild corrosion potential.

Based on the results of our analyses, the soils at the site appear to exhibit a severe tendency to corrode buried steel. A Corrosion Engineer should review the test results discussed herein, when designing appropriate methods of protecting buried steel.

The degradation of concrete is caused by chemical agents in the soil or groundwater that react with concrete to either dissolve the cement paste or precipitate larger compounds which cause cracking and flaking. The concentration of water-soluble sulfates in the soils is a good indicator of the potential for chemical attack of concrete. Sulfate concentrations in soil can be used to evaluate the need for protection of concrete based on American Concrete Institute (ACI) guidelines. General guidelines concerning the potential for degradation of concrete as a function of sulfate ion concentration are presented in the following table.

Aggressiveness	Sulfate Ion Concentration (ppm)
Very severe	> 20,000
Severe	2,000 – 20,000
Moderate	1,000 – 2,000
Mild	< 1,000

The results of sulfate content tests performed on selected soil samples from this study ranged from 26 to 144 ppm which indicate the potential for deterioration of concrete, based on sulfate ion concentrations, is mild. We recommend that a Corrosion Engineer be consulted to determine if protective measures are warranted.

Limitations

The geotechnical engineering recommendations presented herein are based on the geotechnical engineer's experience and professional opinion. These services were performed with the degree of skill and care normally utilized by other members of the geotechnical engineering profession practicing in this location and at this time. There is no warranty, either express or implied. The results, conclusions, and recommendations contained herein are directed at, and intended to be utilized within our contracted scope of work. This memo is not intended to be used for any other purposes.

The analyses, conclusions, and recommendations in this memo are based on the subsurface and depth-to-water conditions present in the borings and the engineering characteristics of the soil as determined through field and laboratory testing at this point in time. The memo does not reflect variations in subsurface conditions that may exist between or beyond these borings. Variations in soil conditions should be expected between the borings, the nature and extent of which may not become evident until construction is undertaken. Subsurface and depth-to-water conditions can change over time due to both natural and manmade forces, including changes in condition and/or use of adjacent properties.

Attachment

Standard Testing and Engineering Company, LLC (STEC) Geotechnical Data Study

SUBSURFACE EXPLORATION

B2280 - Chilled Water System and Water Cooling Tower

Tinker Air Force Base

Oklahoma City, Oklahoma

PROJECT NO. 2230-0534



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October 27, 2022

HDR Engineering, Inc.
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Attn: Mr. Thomas Wesling, P.E.

Re: Subsurface Exploration
B2280 - Chilled Water System and Water Cooling Tower
Tinker Air Force Base
Oklahoma City, Oklahoma

Dear Mr. Wesling:

Standard Testing & Engineering, LLC (Standard Testing) is pleased to present the report covering the subsurface exploration for the subject project. This study was authorized by signed Subconsultant Agreement 10003000002062, dated August 9th, 2022.

Standard Testing conducted a geotechnical investigation at the site of the new chilled water system and water cooling tower project in Tinker Air Force Base, Oklahoma. This report contains the detailed results of the geotechnical investigation, including subsurface conditions and laboratory results.

The subsurface soils consist of approximately 50.8 feet of clays with varying amounts of sand and sands with varying amounts of silt overlying weathered shale, shale, and loosely cemented sandstone and exhibit non-plastic to moderately plastic characteristics.

We trust that the results and recommendations contained herein will permit adequate economical design and construction of the proposed structure. Unless you specify otherwise, we will keep samples obtained from these borings in our Oklahoma City laboratory for the next thirty (30) days.

We appreciate the opportunity to assist on this project. Please call on us if we can be of further service.

Respectfully submitted,
STANDARD TESTING & ENGINEERING, LLC

Cody Latham, E.I.
Staff Geotechnical Engineer

Roy Khalife, P.E.
Geotechnical Engineer

Project No. 2230-0534
Account No. 0230HDR20

SUBSURFACE EXPLORATION

B2280 - Chilled Water System and Water Cooling Tower
Tinker Air Force Base
Oklahoma City, Oklahoma

PROJECT NO. 2230-0534

PREPARED FOR

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Prepared By:



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Reviewed By:



Roy Khalife, P.E.
Geotechnical Engineer

I certify my e-signature for the study entitled "Subsurface Exploration."

Dated 10/27/2022

October 27, 2022

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Section 1 INTRODUCTION

1.1 Authorization

This report presents the results of a subsurface exploration performed by Standard Testing & Engineering, LLC (Standard Testing) in accordance with the proposal (P-2022-207) prepared for Mr. Thomas Wesling, dated July 27th, 2022, and identified as Standard Testing project number 2230-0534. This geotechnical study was authorized by signed Subconsultant Agreement 10003000002062, dated August 9th, 2022.

1.2 Purpose and Scope

A geotechnical investigation was performed for the purpose of (1) determining the subsurface conditions, and (2) evaluating the chemical and plasticity characteristics of the soils

Two (2) exploratory borings (B-1 and B-2) were drilled to a depth 50.8 feet. The boring depths and types of testing were performed according to the scope of work proposed by Standard Testing and accepted by Mr. Wesling. Narrative descriptions of our findings are contained in the body of this report. A site and boring location plan, the boring logs, the soil profile, and a summary sheet of laboratory test results are included in the Appendices of this report.

1.3 Project Location and Description

It is understood that the B2280 - Chilled Water Systems and Cooling Tower project is proposed to be constructed at Tinker Air Force Base in Oklahoma City, Oklahoma.

If the project is not as described or has changed, Standard Testing must be notified in order to reevaluate the recommendations for the project.

Section 2 FIELD EXPLORATION

2.1 Drilling Information

The field exploration work was performed between the 11th and 14th of October, 2022. Conditions at the site were investigated with two (2) borings at the locations indicated on the site and boring location plan, included in Appendix "A." The boring surface elevations were measured with respect to a Temporary Bench Mark (TBM) established at the N.W. corner of the existing building. The Temporary Bench Mark (TBM) location is also shown in the site and boring location plan in Appendix "A." Boring surface elevations, rounded to the nearest foot, are reported on the individual boring logs, included in Appendix "A."

The benchmark has an assigned relative elevation of 100 feet. Boring depths were 50.8 feet within the project site. A total of 10 inches of asphalt pavement and aggregate base were present at both drilling locations and boring and sampling depths were adjusted to account for this. For accurate sampling, cuttings were observed continuously during drilling with specific samples being taken at distinct lithologic changes. The equipment used, field tests performed, and soil samples taken are discussed below.

2.2 Equipment Used

Two (2) borings were drilled with a truck-mounted CME-55 rotary drilling unit equipped with 3.25" I.D. X 7.25" O.D. hollow stem augers (HSA). Standard penetration tests (SPT) used a 1.375" ID split spoon sampler driven by an automatic hammer utilizing a 140 lb. weight falling 30 inches.

2.3 Testing and Sampling Performed

Standard penetration tests were performed in order to estimate the shear strengths of the soils in their natural state. The test was conducted as specified by ASTM D1586, "Penetration Test and Split-Barrel Sampling of Soils." The in-situ bearing strength is related to the N-value from this test. "N" is the number of blows required to drive a split-spoon sampler twelve inches, after a 6-inch seating, into undisturbed soil. The soil samples recovered in the split-spoon barrel were removed from the sample tool in the field, visually classified, and labeled according to boring number and depth. Results of the standard penetration tests are denoted at their respective depths on the boring logs.

Thin-walled tube samples were collected as specified by ASTM D1587, "Standard Practice for Thin-Walled Tube Sampling of Soils for Geotechnical Purposes." Two (2) thin-walled samples were collected for this project, however due to the presence of asphalt debris in the top five (5) feet, one (1) sample was non-viable for unconfined compressive strength testing.

Depths of individual split spoon (standard penetration tests), thin-walled tube, and grab samples are indicated on the boring logs included in Appendix "B." All samples were labeled and sealed in water tight, protective containers and returned to the laboratory for further evaluation and testing.

2.4 Subsurface Conditions

The soils encountered consist of clays with varying amounts of sand and sands with varying amounts of silt overlying weathered shale, shale, and loosely cemented sandstone. The cohesive soils were found to be firm to very stiff in consistency and the cohesionless soils were found to have medium dense relative density. Rock materials (i.e., defined by standard penetration test refusal) were encountered in the indicated borings at the relative elevation shown in the following table:

Table 1: Relative Elevation of Rock Material

Boring No.	Surface Elevation (feet)	Rock Depth (feet)	Rock Elevation (feet)	Rock Material
B-1	100.0	16.8	83.2	Shale Rock
		25.8	74.2	Sandstone Rock
B-2	99.0	15.8	83.2	Shale Rock
		30.8	68.2	Sandstone Rock

2.5 Groundwater

During drilling and at completion of drilling operations, groundwater was NOT encountered in the borings. Water Levels were measured during, at completion, and fifteen minutes after the borings were completed. Presence of water should be anticipated in any excavation. Water travelling through soil (subsurface water) is often unpredictable and may be present at shallow depths. Due to the seasonal changes in groundwater and the unpredictable nature of groundwater paths, groundwater levels will fluctuate. Therefore, it is necessary during construction to be observant for groundwater seepage in excavations in order to assess the situation and make necessary changes. We cannot assume responsibility for difficulties experienced during construction or for future operational problems due to elevation or volume of water encountered.

Section 3

LABORATORY TESTING

Laboratory testing was performed in order to determine the plasticity characteristics of the subsurface materials as well as confirm the soil classifications.

3.1 Tests Performed

- Moisture content tests were performed on split spoon, thin-walled tube, and bag samples, in accordance with ASTM D2216, to determine the in-situ moisture conditions.
- Density tests were performed on intact split spoon, and thin-walled tube samples in accordance with ASTM D7263 Method A.
- Atterberg limits tests were performed on split spoon, thin-walled tube, and bag samples to determine the plasticity characteristics and swell potential of the soil. The tests were performed in accordance with ASTM D4318.
- Sieve analyses were performed on split spoon, thin-walled tube, and bag samples, in accordance with ASTM D2487, for aid in soil classification. These soils were classified according to the Unified Soil Classification System (USCS) and the American Association of State Highway and Transportation Officials (AASHTO) soil classification system.
- Laboratory pH and resistivity tests were conducted on split spoon and grab soil samples. AASHTO T289 was used to determine the pH of the soil, while AASHTO T288 method was employed to determine the electrical resistivity of the sampled soil. pH and resistivity tests determine the corrosion potential of the on-site soils regard to underground structures and pipelines. The test results are summarized under Laboratory Testing Results in Section 4 of this report.
- Soluble sulfate content tests were performed on split spoon and grab soil samples in accordance with OHDL-49. The test results are summarized under Laboratory Testing Results in Section 4 of this report.
- Soluble chloride content tests were performed on split spoon and grab soil samples by colorimetric method using a SMART2 Colorimeter. The test results are summarized under Laboratory Testing Results in Section 4 of this report.
- Unconfined compressive strength testing was conducted on a thin-walled tube soil sample in accordance with ASTM D2166. The unconfined compressive strength as determined by this test, along with the results of the standard penetration test, is used to estimate the in-

situ shear strength of the various soils encountered. The graph in Appendix "D" depicts the behavior of the tested soil under compression without confinement. The unconfined compressive strength of the soil sample is presented on the boring logs and in the "Summary of Laboratory Test Results" table in Appendix "D."

3.2 Laboratory Summary

General descriptions of the encountered soils together with visual and laboratory classifications and numerical values of the test results are on the boring logs and soil profile included in Appendix "B." A "Summary of Test Results" is included in Appendix "D."

Section 4

ENGINEERING EVALUATION AND RECOMMENDATIONS

4.1 Soil Conditions

The soils encountered in this investigation consist of clays with varying amounts of sand and sands with varying amounts of silt overlying weathered shale, shale, and loosely cemented sandstone. The cohesive soils were found to be firm to very stiff in consistency and the cohesionless soils were found to have medium dense relative density. These near surface soils exhibit non-plastic to moderately plastic characteristics. Rock materials (i.e., defined by standard penetration test refusal) were encountered in both borings.

4.2 Laboratory Testing Results

pH and Resistivity Test Results

The pH and resistivity of the soil samples tested are summarized in the following table:

Table 2: pH and Resistivity Content Test Results

Boring No.	Sample I.D.	Depth (feet)	pH	Resistivity (ohm-cm)	Corrosivity
B-1	A	0.8-2.3	7.1	1,622	Moderately
B-2	B	2.3-3.8	8.9	3,611	Slightly
B-2	E	8.8-10.3	7.8	4,919	Slightly

The pH and resistivity test results indicate that soils are slightly to moderately corrosive with regards to accelerated corrosion of metals.

Soluble Sulfate and Chloride Content Test Results

The soluble sulfate and chloride test results are included in the following table:

Table 3: Soluble Sulfate and Chloride Content Test Results

Boring No.	Sample I.D.	Depth (feet)	Sulfate Content (ppm)	Chloride Content (ppm)
B-1	A	0.8-2.3	144	14
B-2	B	2.3-3.8	26	11
B-2	E	8.8-10.3	27	25

The results of the sulfate and chloride content tests indicate that Type I/II cement may be used for concrete and the sulfate levels are negligible in affecting lime stabilization of the soil. According to the National Lime Association's publication entitled "Technical Memorandum:

Guidelines for Stabilization of Soils Containing Sulfates,” which can be found on their website at www.lime.org/documents/publications/free_downloads/technical-memorandum.pdf, If the total level of soluble sulfates is below 0.3%, or 3,000 parts per million (ppm), by weight of soil, then lime stabilization should not be of significant concern. The potential for a harmful reaction is low.”

Unconfined Compressive Strength of Soil Test Results

Unconfined compressive strength test was conducted on thin-walled tube soil sample in accordance with ASTM D2166 testing method. The results are presented in the following table and are also presented in Appendix “D.”

Table 4: Unconfined Compressive Strength Test Results

Boring No.	Depth (feet)	Moisture Content (%)	Dry Density (pcf)	Max. Stress, q_u (psf)	Strain at Max. Stress (%)
B-1	3.8-5.8	9.5	123.7	7,475	3.1

Section 5

BASIS FOR RECOMMENDATIONS

5.1 General Comments

The recommendations and conclusions contained in this report are based on the borings drilled and tests performed. We would point out that there may be variations in material properties over the site and would caution that there may be unknown conditions in existence which differ seriously from those encountered by the test borings. Such conditions, if indeed they exist at all, cannot be, and have not been, accounted for in this report. Therefore, the descriptions, recommendations, and conclusions contained herein should be considered as generalized, applying only to the immediate vicinity of the borings.

5.2 Limitations

Since this report is being prepared in advance of much of the detailed design, the finalized soil and structure parameters (i.e., floor elevation, structural system and loading, vertical movement tolerance, etc.) may differ from the ones considered during the preparation of this report. If such a design variance is substantial, Standard Testing would request the opportunity to review the plans and specifications of the proposed facility for applicability to the soil conditions in this report, and assurance of consistency with its intent.

It is recommended that Standard Testing be retained for testing and observation during earthwork and foundation construction phases, to help determine that the design requirements are fulfilled.

This report has been prepared for the exclusive use of our client for specific application to the project discussed and has been prepared in accordance with generally accepted geotechnical practice.

APPENDIX A

Vicinity Map Site and Boring Location Plan



Vicinity Map

Project Name: B2280 Chilled Water System & Cooling Towers
Project Location: Tinker Air Force Base, Oklahoma
Project No.: 2230-0534

STANDARD
TESTING
AND ENGINEERING COMPANY



Site and Boring Location Plan

Project Name: B2280 Chilled Water System & Cooling Towers
Project Location: Tinker Air Force Base, Oklahoma
Project No.: 2230-0534

APPENDIX B

Boring Logs

Soil Profile

Definition of Descriptive Terms

BORING LOG B-1

(1 of 2)

PROJECT NAME: B2280 Chilled Water System & Cooling
PROJECT NUMBER: 2230-0534
PROJECT LOCATION: Tinker Air Force Base
CLIENT: HDR Engineering, Inc.

DEPTH (FT) ELEVATION (FT)	GRAPHIC LOG	USCS	SAMPLER SYMBOLS			SAMPLE TYPE	SAMPLE NUMBER	BLOW COUNTS (N)	POCKET PENTROMETER (tsf)	RECOVERY % / RQD	WATER CONTENT (%)	DRY UNIT WEIGHT (pcf)	UCS (psf)	#200 SIEVE (%)	ATTEBERG LIMITS	
			 Grab	 ST	 RC											
			 SS	 TC	 HA											
			MATERIAL DESCRIPTION													
0			4" (ASPHALT)													
			6" (AGGREGATE)				A									
		CL	Reddish Brn. SANDY CLAY (W/ ASPHALT DEBRIS)				B	7-8-10 (18)		56	12.8			91.9	38-14-24	
96			Reddish Brn. LEAN CLAY (W/ ASPHALT DEBRIS)				C			100	11.7	124	7475			
6		CL	Moist, Moderate Plasticity, V. Stiff (W/O ASPHALT DEBRIS)				D	6-12-13 (25)		72	9.7			86.8	28-14-14	
			Sl. Moist, Low Plasticity													
		CL	Reddish Brn. WEATHERED SHALE V. Soft Rock				E	4-16-31 (47)		75						
90			Sl. Moist, Moderate Plasticity				F	8-17-20 (37)		83	12.3			95.9	34-15-19	
12																
							G	8-40-50/1.00"		100						
18		CL	(ROCK) Reddish Brn. SHALE Med. Hard Rock													
			Sl. Moist, Moderate Plasticity				H	19-45-50/2.00"		100	11.0			85.4	32-13-19	
78																
24																
			Reddish Brn. LOOSELY CEMENTED SANDSTONE Soft Rock				I	50/3.00"		100						
72																
30							J	50/4.00"		100						
66																
36							K	50/3.50"		100						

WATER LEVELS			ELEVATIONS / LOCATIONS		DRILLING			
WD	▽	-	GROUND ELEVATION: 100		DRILL START:	10/12/2022	LOGGER:	J.B.
AD	▽	-	TBM: N.W. Corner of Building		DRILLED END:	10/13/2022	DRILLER:	G.D.
24 Hrs	▽	-	GPS: 35 24.793'N, 97 22.564'W		DRILL RIG:	CME-55	HOLE SIZE:	7.25"
> 24 Hrs	▽	-	STA:	OFFSET:	DRILL METHOD:	HSA		

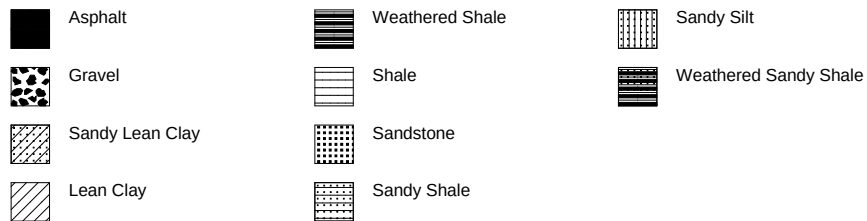
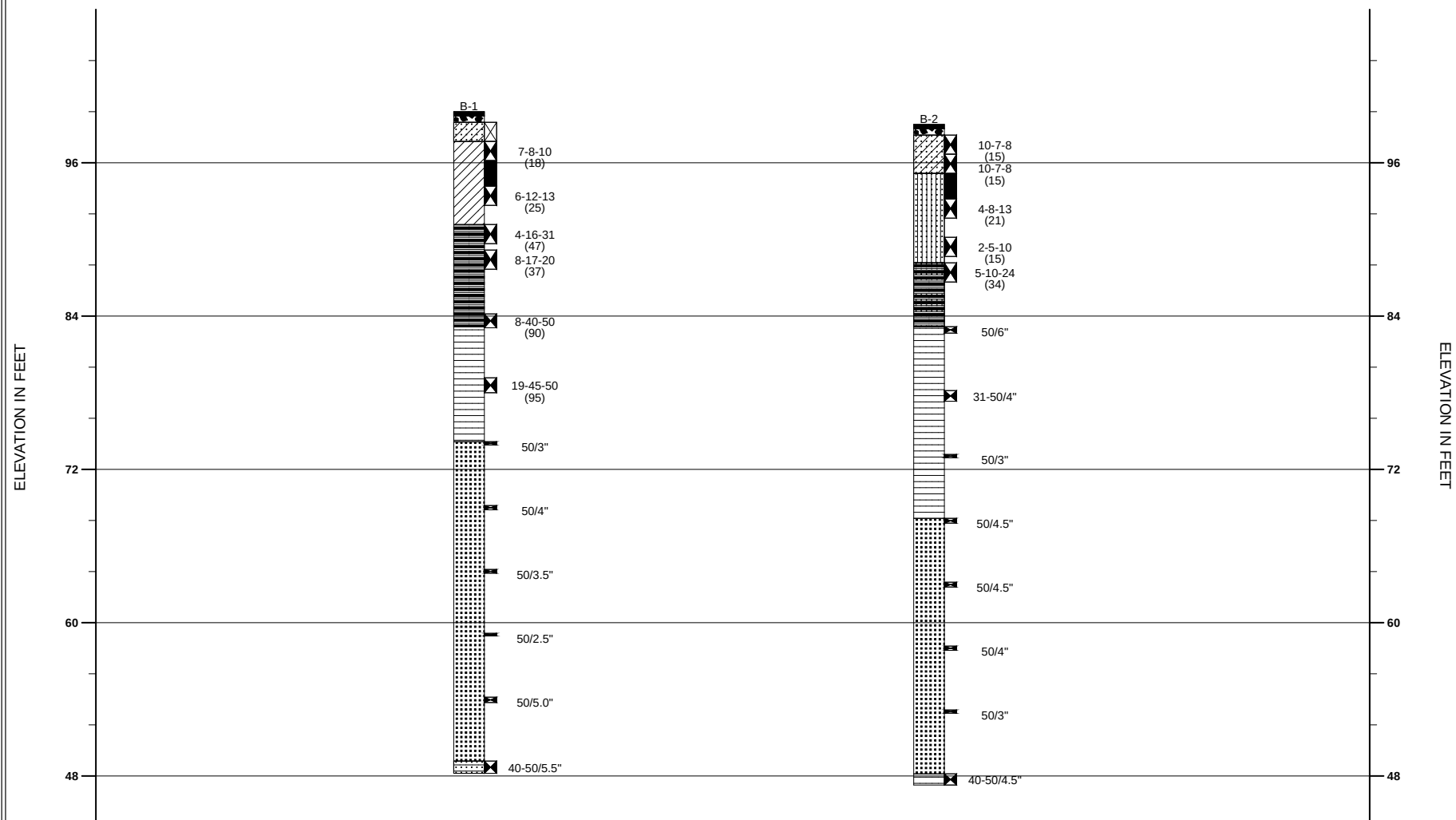
BORING LOG B-2

(1 of 2)

PROJECT NAME: B2280 Chilled Water System & Cooling
PROJECT NUMBER: 2230-0534
PROJECT LOCATION: Tinker Air Force Base
CLIENT: HDR Engineering, Inc.

DEPTH (FT) ELEVATION (FT)	GRAPHIC LOG	USCS	SAMPLER SYMBOLS			SAMPLE TYPE	SAMPLE NUMBER	BLOW COUNTS (N)	POCKET PENETROMETER (tsf)	RECOVERY % / RQD	WATER CONTENT (%)	DRY UNIT WEIGHT (pcf)	UCS (psf)	#200 SIEVE (%)	ATTERBERG LIMITS	
			 Grab	 ST	 RC											
			 SS	 TC	 HA											
			MATERIAL DESCRIPTION													
0			4" (ASPHALT)													
			6" (AGGREGATE)				A	10-7-8 (15)		78	15.6			53.4	31-15-16	
96		CL	Reddish Brn. SANDY LEAN CLAY (W/ ASPHALT DEBRIS)				B	10-7-8 (15)		78						
			Moist, Moderate Plasticity, Stiff													
6		SM	Reddish Brn. SILTY SAND (W/ ASPHALT DEBRIS)				C			100	14.7			28.1	NP-NP-NP	
		SM	V. Moist, Nonplastic (W/O ASPHALT DEBRIS)				D	4-8-13 (21)		89	15.5			24.3	16-13-3	
			Med. Dense, Low Plasticity													
90							E	2-5-10 (15)		100						
12		CL	Reddish Brn. WEATHERED SANDY SHALE				F	5-10-24 (34)		100	17.9			52.6	26-11-15	
			V. Moist, Moderate Plasticity, V. Soft Rock													
84																
18			(ROCK) Reddish Brn. SHALE Soft Rock				G	50/6.00"		100						
78							H	31-50/4.00"		100						
24																
							I	50/3.00"		100						
72			Med. Hard Rock													
30							J	50/4.50"		100						
			Reddish Brn. LOOSELY CEMENTED SANDSTONE													
66			Soft Rock				K	50/4.50"		100						
36																

WATER LEVELS			ELEVATIONS / LOCATIONS		DRILLING			
WD	▽	-	GROUND ELEVATION: 99		DRILL START:	10/11/2022	LOGGER:	T.L.
AD	▽	-	TBM: N.W. Corner of Building		DRILLED END:	10/14/2022	DRILLER:	G.D.
24 Hrs	▽	-	GPS: 35 24.808'N, 97 22.565'W		DRILL RIG:	CME-55	HOLE SIZE:	7.25"
> 24 Hrs	▽	-	STA:	OFFSET:	DRILL METHOD:	HSA		



SOIL PROFILE

B2280 Chilled Water System & Cooling Towers

PROJECT NO. 2230-0534

HORIZONTAL

VERTICAL

SCALE:

SCALE: 1"=12'

DEFINITION OF DESCRIPTIVE TERMS

Consistency of Cohesive Soils (at moisture content near plastic limit):

- Very Soft - Easily penetrated 4" to 6" by fist; tall core will sag under its own weight.
- Soft - Easily molded by fingers.
- Firm - Can be penetrated 2" to 3" by thumb with moderate effort, imprinted with fingers.
- Stiff - Readily indented by thumb but penetrated only with great effort.
- Very Stiff - Readily indented by thumbnail, imprinted very slightly with pressure from fingers.
- Hard - Indented with difficulty by thumbnail, cannot be imprinted with fingers.

Density of Cohesionless Soils:

- Very Loose - less than 4 SPT "N" value corrected for overburden.
- Loose - 5 to 10 SPT "N" value corrected for overburden.
- Medium Dense - 11 to 30 SPT "N" value corrected for overburden.
- Dense - 31 to 50 SPT "N" value corrected for overburden.
- Very Dense - 51 to 50/6" SPT "N" value corrected for overburden.
- Hard - less than 6" penetration in 50 SPT "N" blows corrected for overburden (cemented).

Hardness of Rock:

- Very Soft - can be scratched readily by fingernail
- Soft - can be grooved readily by knife or pick
- Medium - can be grooved 0.05" deep by firm pressure of knife
- Moderately Hard - can be scratched by knife
- Hard - can be scratched by knife or pick only with difficulty
- Very Hard - cannot be scratched by knife or sharp pick

Other Terms Descriptive of Consistency:

- Brittle - Ruptures with little deformation
- Friable - Crumbles or pulverizes easily.
- Elastic - Returns to original length after small deformation.
- Spongy - Is very porous, loose and elastic.
- Sticky - Adheres or sticks to tools or hands.

In-Situ Moisture Descriptions:

- Dry - powdery
- Slightly Moist - water not readily absorbed by paper
- Moist - water readily absorbed by paper
- Very Moist - water condenses on sample tray
- Wet - water drips from sample

Degree of Plasticity When Moist to Very Moist:

- Nonplastic - cannot be rolled into a ball
- Trace of Plasticity - can be rolled into a ball but not into a 1/8" thread
- Low Plasticity - barely holds its shape when rolled into a 1/8" thread
- Fairly Low Plasticity - 1/8" thread quickly ruptures when bent
- Medium Plasticity - 1/8" thread withstands considerable deformation without rupture.
- Fairly High Plasticity - difficult to rupture a 1/8" thread by bending.
- High Plasticity - can be kneaded without rupture; greasy texture.

Abbreviations:

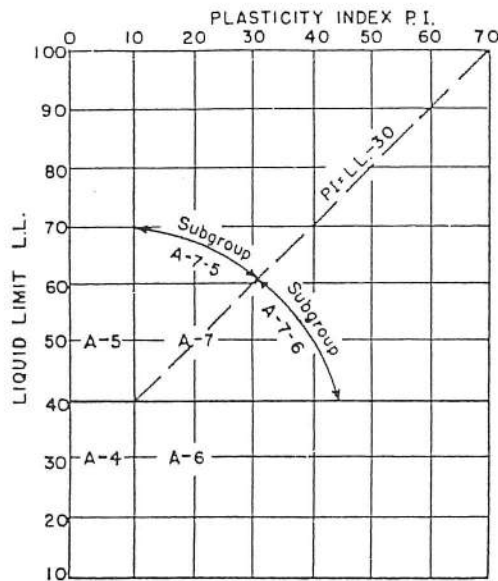
- | | | |
|----------------|---------------|--------------|
| V. - Very | Dk. - Dark | Blk. - Black |
| Tr. - Trace | Lt. - Light | Brn. - Brown |
| Fl. - Fairly | Med. - Medium | |
| Sl. - Slightly | | |

APPENDIX C

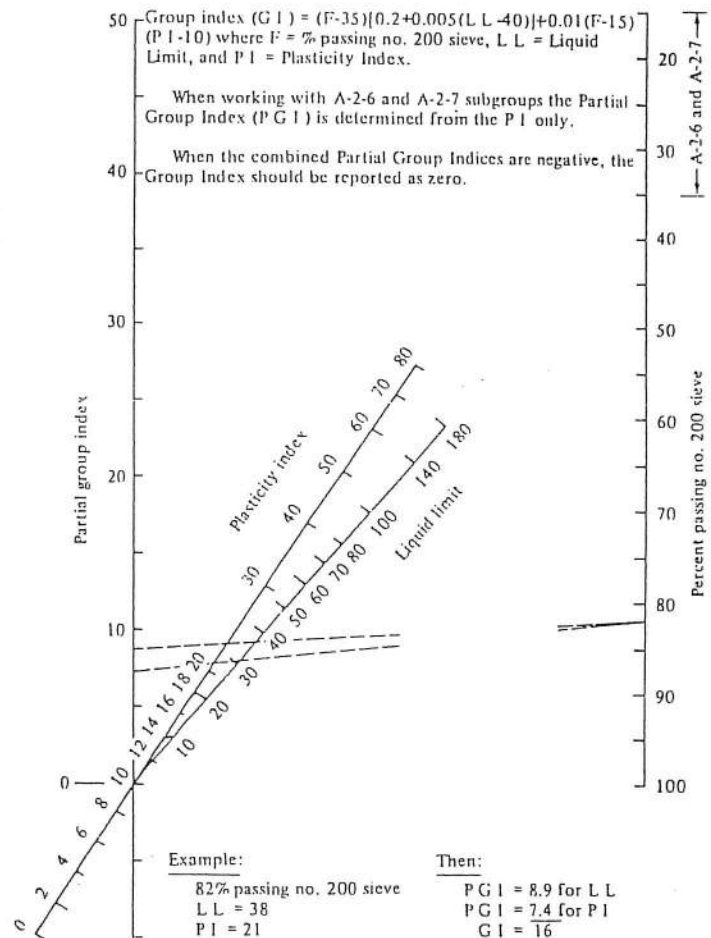
**AASHTO Soil Classification System
Unified Soil Classification System**

Soil Classification System — American Association of State Highway and Transportation Officials

The tables and charts given below are from AASHTO Designation: M 145-83, The Classification of Soils and Soil-Aggregate Mixtures for Highway Construction Purposes. More detailed information as to the background and application of the system may be obtained from the report.



Liquid-limit and plasticity-index ranges for the A-4, A-5, A-6 and A-7 subgrade groups.



Group index chart

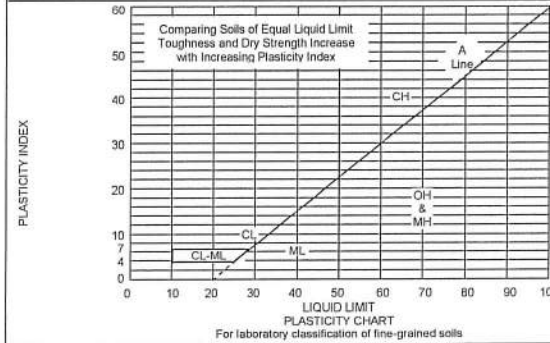
Classification of Soils and Soil-Aggregate Mixtures (with Suggested Subgroups)

General classification	Granular materials (35 per cent or less passing No. 200)							Silt-clay materials (More than 35 per cent passing No. 200)			
Group classification	A-1		A-3	A-2				A-4	A-5	A-6	A-7
	A-1-a	A-1-b		A-2-4	A-2-5	A-2-6	A-2-7				A-7-5; A-7-6
Sieve analysis; Per cent passing: No. 10 No. 40 No. 200	50 max. 30 max. 15 max.	— 50 max. 25 max.	— 51 min. 10 max.	— — 35 max.	— — 35 max.	— — 35 max.	— — 35 max.	— — 36 min.	— — 36 min.	— — 36 min.	— — 36 min.
Characteristics of fraction passing No. 40; Liquid limit Plasticity index	— 6 max.		— NP	40 max. 10 max.	41 min. 10 max.	40 max. 11 min.	41 min. 11 min.	40 max. 10 max.	41 min. 10 max.	40 max. 11 min.	41 min. 11 min.*
Usual types of significant constituent materials	Stone fragments, gravel and sand		Fine sand	Silty or clayey gravel and sand				Silty soils		Clayey soils	
General rating as subgrade	Excellent to good						Fair to poor				

*P.I. of A-7-5 subgroup is equal to or less than L.L. minus 30. P.I. of A-7-6 subgroup is greater than L.L. minus 30

UNIFIED SOIL CLASSIFICATION (Including Identification and Description)														
Major Divisions		Group Symbols	Typical Names	Field Identification Procedures (Excluding particles larger than 3 inches and basing fractions on estimated weights)		Information Required for Describing Soils	Laboratory Classification Criteria							
1	2	3	4	5		6	7							
Coarse-grained Soils More than half of material is larger than No. 200 sieve size. The No. 200 sieve size is about the smallest particle visible to the naked eye.	Gravels More than half of coarse fraction is larger than No. 4 sieve size. Sands More than half of coarse fraction is smaller than No. 4 sieve size. (For Visual classification, the 1/4-in size may be used as equivalent to the No. 4 sieve size)	Clean Gravels (Little or no fines)	GW	Well-graded gravels, gravel-sand mixtures, little or no fines.	Wide range in grain sizes and substantial amounts of all intermediate particle sizes.	For undisturbed soils add information on stratification, degree of compactness, cementation, moisture conditions and drainage characteristics Give typical name; indicate approximate percentages of sand and gravel, maximum size, angularity, surface condition, and hardness of the coarse grains, local or geologic name and other pertinent descriptive information; and symbol in parentheses. Example: Silty sand, gravelly; about 20% hard, angular gravel particles 1/4-in. maximum size; rounded and subangular sand grains coarse to fine, about 15% nonplastic fines with low dry strength; well compacted and moist in place; alluvial sand; (SM).	Determine percentages of gravel and sand from grain size curve. Depending on percentage of fines (fraction smaller than No. 200 sieve size) coarse-grained soils are classified as follows: Less than 5% More than 12% 5% to 12%	$C_u = \frac{D_{60}}{D_{10}}$ Greater than 4 $C_c = \frac{(D_{30})^2}{D_{10} \times D_{60}}$ Between 1 and 3 Not meeting all gradation requirements for GW						
			GP	Poorly-graded gravels, gravel-sand mixtures, little or no fines.	Predominantly one size or a range of sizes with some intermediate sizes missing.			Atterberg limits below "A" line or PI less than 4						
		Gravels with Fines (Appreciable amount of fines)	GM	Silty gravels, gravel-sand-silt mixtures.	Nonplastic fines or fines with low plasticity (for identification procedures see ML below).			Atterberg limits above "A" line or PI greater than 7						
			GC	Clayey gravels, gravel-sand-clay mixtures.	Plastic fines (for identification procedures see CL below).			Above "A" Line with PI between 4 and 7 are borderline cases requiring use of dual symbols						
		Clean Sands (Little or no fines)	SW	Well-graded sands, gravelly sands, little or no fines.	Wide range in grain size and substantial amounts of all intermediate particle sizes.			$C_u = \frac{D_{60}}{D_{10}}$ Greater than 6 $C_c = \frac{(D_{30})^2}{D_{10} \times D_{60}}$ Between 1 and 3 Not meeting all gradation requirements for SW						
			SP	Poorly-graded sands, gravelly sands, little or no fines.	Predominantly one size or a range of sizes with some intermediate sizes missing.			Atterberg limits below "A" line or PI less than 4						
	Sands with Fines (Appreciable amount of fines)	Sands with Fines (Appreciable amount of fines)	SM	Silty sands, sand-silt mixtures.	Nonplastic fines or fines with low plasticity (for identification procedures see ML below).			Atterberg limits above "A" line or PI greater than 7						
			SC	Clayey sands, sand-clay mixtures.	Plastic fines (for identification procedures see CL below).			Limits plotting in hatched zone with PI between 4 and 7 are borderline cases requiring use of dual symbols						
			Identification Procedures On Fractions Smaller than No. 40 Sieve Sizes											
				Dry Strength (Crushing characteristics)	Dilatancy (Reaction to shaking)	Toughness (Consistency near PL)								
Fine-grained Soils More than half of material is smaller than No. 200 sieve size. The No. 200 sieve size is about the smallest particle visible to the naked eye.	Silt and Clays Liquid limit less than 50 Silt and Clays Liquid limit greater than 50	Silt and Clays Liquid limit less than 50 Silt and Clays Liquid limit greater than 50	ML	Inorganic silts and very fine sands, rock flour, silty or clayey fine sands or clayey silts with slight plasticity.	None to slight	Quick to slow	None	Give typical name, indicate degree and character of plasticity, amount and maximum size of coarse grains, color in wet condition, odor if any, local or geologic name, and other pertinent descriptive information; and symbol in parentheses. For undisturbed soils add information on structure, stratification, consistency in undisturbed and remolded states, moisture and drainage conditions. Example: Clayey silt, brown, slightly plastic, small percentage of fine sand, numerous vertical root holes, firm and dry in place, loess, (ML).						
			CL	Inorganic clays of low to medium plasticity, gravelly clays, sandy clays, silty clays, lean clays.	Medium to high	None to very slow	Medium							
			OL	Organic silts and organic silty clays of low plasticity.	Slight to medium	Slow	Slight							
			MH	Inorganic silts, micaceous or dolomaceous fine sandy or silty soils, elastic silts.	Slight to medium	Slow to none	Slight to Medium							
			CH	Inorganic clays of high plasticity, fat clays.	High to very high	None	High							
			OH	Organic clays of medium to high plasticity, organic silts.	Medium to high	None to very slow	Slight to medium							
			Highly Organic Soils			PI	Peat and other high organic soils.			Readily identified by color, odor, spongy feel and frequently by fibrous texture.				
			(1) Boundary Classifications: Soils possessing characteristics of two groups are designated by combinations of group symbols. For example GW-GC, well-graded gravel-sand mixture with clay binder. (2) All sieve sizes on this chart are U.S. Standard											
			FIELD IDENTIFICATION PROCEDURES FOR FINE-GRAINED SOILS OR FRACTIONS											
			These procedures are to be performed on the minus No. 40 sieve size particles, approximately 1/64 in. For field classification purposes, screening is not intended, simply remove by hand the coarse particles that interfere with the tests.											
Dilatancy (Reaction to shaking)			Dry Strength (Crushing Characteristics)			Toughness (Consistency near plastic limit)								
After removing particles larger than No. 40 sieve size, prepare a pat of moist soil with a volume of about one-half cubic inch. Add enough water if necessary to make the soil soft but not sticky. Place the pat in the open palm of one hand and shake horizontally, striking vigorously against the other hand several times. A positive reaction consists of the appearance of water on the surface of the pat, which changes to a livery consistency and becomes glossy. When the sample is squeezed between the fingers, the water and glass disappear from the surface, the pat stiffens, and finally it cracks or crumbles. The rapidity of appearance of water during shaking and of its disappearance during squeezing assist in identifying the character of the fines in a soil.			High dry strength is characteristic for clays of the CH group. A typical inorganic silt possesses only very slight dry strength. Silty fine sands and silts have about the same slight dry strength, but can be distinguished by the feel when powdering the dried specimen. Fine sand feels gritty, whereas a typical silt has the smooth feel of flour.			After removing particles larger than the No. 40 sieve size, a specimen of soil about one-half inch cube in size is molded to the consistency of putty. If too dry, water must be added and if sticky, specimen should be spread out in a thin layer and allowed to lose some moisture by evaporation. Then the specimen is rolled out by hand on a smooth surface or between the palms into a thread about one-eighth inch in diameter. The thread is then folded and rerolled repeatedly. During this manipulation the moisture content is gradually reduced and the specimen stiffens, finally loses its plasticity, and crumbles when the plastic limit is reached.								
Very fine clean sands give the quickest and most distinct reactions whereas a plastic clay has no reaction. Inorganic silts, such as a typical rock flour show a moderately quick reaction.						After the thread crumbles, the pieces should be lumped together and slight kneading action continued until lump crumbles.								
						The tougher the thread near the plastic limit and the stiffer the lump when it finally crumbles, the more potent is the colloidal clay fraction in the soil. Weakness of the thread at plastic limit and quick loss of coherence of the lump below the plastic limit indicate either inorganic clay of low plasticity, or materials such as kaolin-type clays and organic clays which occur below the A-line.								
						Highly organic clays have a very weak and spongy feel at the plastic limit.								

Adopted by Corps of Engineers and Bureau of Reclamation, January 1952

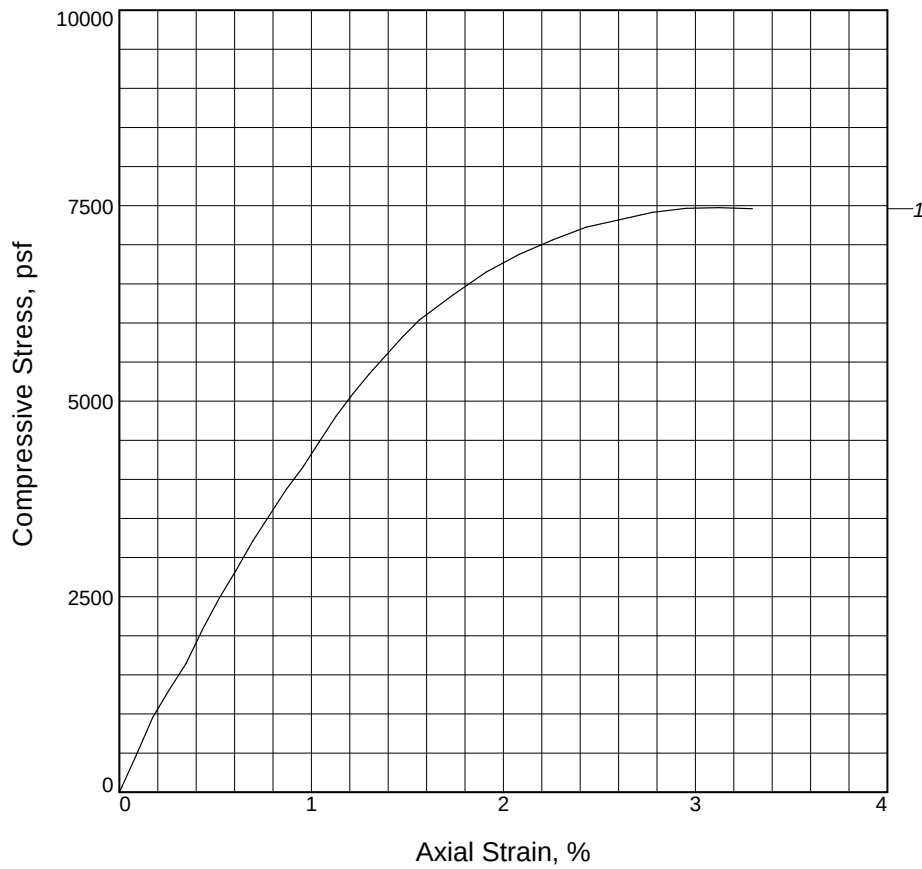


APPENDIX D

Summary of Laboratory Test Results

SUMMARY OF LABORATORY TEST RESULTS

UNCONFINED COMPRESSION TEST



Stage	1			
Unconfined strength, psf	7475			
Undrained shear strength, psf	3738			
Failure strain, %	3.1			
Strain rate, in./min.	0.052			
Water content, %	9.5			
Wet density, pcf	135.5			
Dry density, pcf	123.7			
Saturation, %	74.6			
Void ratio	0.3372			
Specimen diameter, in.	2.87			
Specimen height, in.	5.76			
Height/diameter ratio	2.01			

Description:

LL = PL = PI = Assumed GS= 2.65 Type: Shelby Tube

Project No.: 2230-0534

Date Sampled:

Remarks:

Client: HDR Engineering, Inc.

Project: B2280 Chilled Water System & Cooling Towers

Source of Sample: B-1 Depth: 3.83

Sample Number: C



Figure _____

Area Offices

902 Trails West Loop
202 SE "J" Ave.
7648 E. 46th Place

Enid, OK 73703 (580) 237-3130
Lawton, OK 73501 (580) 353-0872
Tulsa, OK 74145 (918) 289-0005

Report On: Determining Soluble Chloride Content in Soil

Lab No: 1967

Report No: 1967-2

File ID: 2230-0534

Acct ID: 0230HDR20

Page 1 of 1

Client: HDR Engineering, Inc.
Mr. Scott Goehri, P.E.
4435 Main Street, Suite 1000
Kansas City, MO 64111

Project: B2280 Chilled Water System & Water
Cooling Towers
OKC OK

Report Date: 10/27/2022

Location: B-1 (0.8'-2.3')

Sample Date: 10/26/2022

Sampled By:

Field ID: B-1

Test Method: Colorimetric
Material Represented: B-1 (0.8'-2.3')

Blank Adjustment (R'): 0.03 ppm

Mass of Air-Dried Sample Pulverized to #50 Sieve (M_s): 10.12 g

Mass of Sample plus Deionized Water (M_t): 140.26 g

Average Colorimeter Reading on Original Slurry (R): 1.07 ppm

Concentration in Original Solid Sample, C_s = M_t / M_s * (R - R'): ppm

Dilution, if necessary

Mass of Original Slurry Used (M₁): N/A g

Mass of Original Slurry Used plus Additional Deionized Water (M₂): N/A g

Colorimeter Reading on Diluted Slurry (R₁): N/A ppm

Dilution Factor (D_f = M₂ / M₁): No Dilution

Concentration in Original Solid Sample (C_s = D_f * M_t / M_s * [R - R']): ppm

Chloride Concentration in Air-Dry Soil, (C_s): 14 ppm

Test Methods: Colorimetric

Orig: HDR Engineering, Inc. (Kansas City, MO)
Attn: Mr. Scott Goehri, P.E. (1-cc copy)
1-cc Laboratory

Respectfully Submitted,
Standard Testing & Engineering Company

Area Offices

902 Trails West Loop
202 SE "J" Ave.
7648 E. 46th Place

Enid, OK 73703 (580) 237-3130
Lawton, OK 73501 (580) 353-0872
Tulsa, OK 74145 (918) 289-0005

Report On: Determining Soluble Chloride Content in Soil

Lab No: 1967-1

Report No: 1967-1-2

File ID: 2230-0534

Acct ID: 0230HDR20

Page 1 of 1

Client: HDR Engineering, Inc.
Mr. Scott Goehri, P.E.
4435 Main Street, Suite 1000
Kansas City, MO 64111

Project: B2280 Chilled Water System & Water
Cooling Towers
OKC OK

Report Date: 10/27/2022

Location: B-2 (2.3'-3.8')

Sample Date: 10/26/2022

Sampled By:

Field ID: B-2

Test Method: Colorimetric
Material Represented: B-2 (2.3'-3.8')

Blank Adjustment (R'): 0.03 ppm

Mass of Air-Dried Sample Pulverized to #50 Sieve (M_s): 10.18 g

Mass of Sample plus Deionized Water (M_t): 140.37 g

Average Colorimeter Reading on Original Slurry (R): 0.8 ppm

Concentration in Original Solid Sample, C_s = M_t / M_s * (R - R'): ppm

Dilution, if necessary

Mass of Original Slurry Used (M₁): N/A g

Mass of Original Slurry Used plus Additional Deionized Water (M₂): N/A g

Colorimeter Reading on Diluted Slurry (R₁): N/A ppm

Dilution Factor (D_f = M₂ / M₁): No Dilution

Concentration in Original Solid Sample (C_s = D_f * M_t / M_s * [R - R']): ppm

Chloride Concentration in Air-Dry Soil, (C_s): 11 ppm

Test Methods: Colorimetric

Orig: HDR Engineering, Inc. (Kansas City, MO)
Attn: Mr. Scott Goehri, P.E. (1-cc copy)
1-cc Laboratory

Respectfully Submitted,
Standard Testing & Engineering Company

Area Offices

902 Trails West Loop
202 SE "J" Ave.
7648 E. 46th Place

Enid, OK 73703 (580) 237-3130
Lawton, OK 73501 (580) 353-0872
Tulsa, OK 74145 (918) 289-0005

Report On: Determining Soluble Chloride Content in Soil

Lab No: 1967-2

Report No: 1967-2-2

File ID: 2230-0534

Acct ID: 0230HDR20

Page 1 of 1

Client: HDR Engineering, Inc.
Mr. Scott Goehri, P.E.
4435 Main Street, Suite 1000
Kansas City, MO 64111

Project: B2280 Chilled Water System & Water
Cooling Towers
OKC OK

Report Date: 10/27/2022

Location: B-2 (8.8'-10.3')

Sample Date: 10/26/2022

Sampled By:

Field ID: B-2

Test Method: Colorimetric

Material Represented: B-2 (8.8'-10.3')

Blank Adjustment (R'): 0.03 ppm

Mass of Air-Dried Sample Pulverized to #50 Sieve (M_s): 10.18 g

Mass of Sample plus Deionized Water (M_t): 140.28 g

Average Colorimeter Reading on Original Slurry (R): 1.87 ppm

Concentration in Original Solid Sample, C_s = M_t / M_s * (R - R'): ppm

Dilution, if necessary

Mass of Original Slurry Used (M₁): N/A g

Mass of Original Slurry Used plus Additional Deionized Water (M₂): N/A g

Colorimeter Reading on Diluted Slurry (R₁): N/A ppm

Dilution Factor (D_f = M₂ / M₁): No Dilution

Concentration in Original Solid Sample (C_s = D_f * M_t / M_s * [R - R']): ppm

Chloride Concentration in Air-Dry Soil, (C_s): 25 ppm

Test Methods: Colorimetric

Orig: HDR Engineering, Inc. (Kansas City, MO)
Attn: Mr. Scott Goehri, P.E. (1-cc copy)
1-cc Laboratory

Respectfully Submitted,
Standard Testing & Engineering Company

Area Offices

902 Trails West Loop
202 SE "J" Ave.
7648 E. 46th Place

Enid, OK 73703
Lawton, OK 73501
Tulsa, OK 74145

(580) 237-3130
(580) 353-0872
(918) 289-0005

Report On: pH and Resistivity of Soil

Lab No: 1967

Report No: 1967

File ID: 2230-0534

Acct ID: 0230HDR20

Page 1 of 1

Client: HDR Engineering, Inc.
Mr. Scott Goehri, P.E.
4435 Main Street, Suite 1000
Kansas City, MO 64111

Project: B2280 Chilled Water System & Water
Cooling Towers
OKC OK

Report Date: 10/27/2022

Location: B-1 (0.8'-2.3')

Sample Date: 10/26/2022

Sampled By:

Field ID: B-1

Material: B-1 (0.8'-2.3')

pH Test (AASHTO T-289)

pH reading of sample in water	7.1
-------------------------------	-----

Resistivity Test (ASTM G57)

Resistance of Soil in Soil Box, Ohms	1550
Soil Box Geometric Factor	1.0467 cm
Resistivity of Soil, Ohms * cm	1622

Interpreted Corrosivity:

Moderately Corrosive

Test Methods: AASHTO T-289 & ASTM G57

Orig: HDR Engineering, Inc. (Kansas City, MO)
Attn: Mr. Scott Goehri, P.E. (1-cc copy)
1-cc Laboratory

Respectfully Submitted,
Standard Testing & Engineering Company

Area Offices

902 Trails West Loop
202 SE "J" Ave.
7648 E. 46th Place

Enid, OK 73703
Lawton, OK 73501
Tulsa, OK 74145

(580) 237-3130
(580) 353-0872
(918) 289-0005

Report On: pH and Resistivity of Soil

Lab No: 1967-1

Report No: 1967-1

File ID: 2230-0534

Acct ID: 0230HDR20

Page 1 of 1

Client: HDR Engineering, Inc.
Mr. Scott Goehri, P.E.
4435 Main Street, Suite 1000
Kansas City, MO 64111

Project: B2280 Chilled Water System & Water
Cooling Towers
OKC OK

Report Date: 10/27/2022

Location: B-2 (2.3'-3.8')

Sample Date: 10/26/2022

Sampled By:

Field ID: B-2

Material: B-1 (2.3'-3.8')

pH Test (AASHTO T-289)

pH reading of sample in water	8.9
-------------------------------	-----

Resistivity Test (ASTM G57)

Resistance of Soil in Soil Box, Ohms	3450
Soil Box Geometric Factor	1.0467 cm
Resistivity of Soil, Ohms * cm	3611

Interpreted Corrosivity:

Slightly Corrosive

Test Methods: AASHTO T-289 & ASTM G57

Orig: HDR Engineering, Inc. (Kansas City, MO)
Attn: Mr. Scott Goehri, P.E. (1-cc copy)
1-cc Laboratory

Respectfully Submitted,
Standard Testing & Engineering Company

Area Offices

902 Trails West Loop
202 SE "J" Ave.
7648 E. 46th Place

Enid, OK 73703
Lawton, OK 73501
Tulsa, OK 74145

(580) 237-3130
(580) 353-0872
(918) 289-0005

Report On: pH and Resistivity of Soil

Lab No: 1967-2

Report No: 1967-2

File ID: 2230-0534

Acct ID: 0230HDR20

Page 1 of 1

Client: HDR Engineering, Inc.
Mr. Scott Goehri, P.E.
4435 Main Street, Suite 1000
Kansas City, MO 64111

Project: B2280 Chilled Water System & Water
Cooling Towers
OKC OK

Report Date: 10/27/2022

Location: B-2 (8.8'-10.3')

Sample Date: 10/26/2022

Sampled By:

Field ID: B-2

Material: B-2 (8.8'-10.3')

pH Test (AASHTO T-289)

pH reading of sample in water	7.8
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Resistivity Test (ASTM G57)

Resistance of Soil in Soil Box, Ohms	4700
Soil Box Geometric Factor	1.0467 cm
Resistivity of Soil, Ohms * cm	4919

Interpreted Corrosivity:

Slightly Corrosive

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Report On: Determining Soluble Sulfate Content in Soil

Lab No: 1967

Report No: 1967-1

File ID: 2230-0534

Acct ID: 0230HDR20

Page 1 of 1

Client: HDR Engineering, Inc.
Mr. Scott Goehri, P.E.
4435 Main Street, Suite 1000
Kansas City, MO 64111

Project: B2280 Chilled Water System & Water
Cooling Towers
OKC OK

Report Date: 10/27/2022

Location: B-1 (0.8'-2.3')

Sample Date: 10/26/2022

Sampled By:

Field ID: B-1

Test Method: OHD L-49

Material Represented: B-1 (0.8'-2.3')

Mass of Air-Dried Sample Passing #10 Sieve (W_s): 5.09 g

Mass of Deionized Water Used in Slurry (W_w): 200.06 g

Average Colorimeter Reading (R): 3.67 ppm

Volume of Original Filtrate Used (V_f): N/A ml

Volume of Deionized Water Added to Filtrate (V_w): N/A ml

Colorimeter Reading on Diluted Filtrate (R_d): N/A ppm

$$C = [R * (W_w / W_s)] \text{ or } [R_d * (W_w / W_s) * (V_f + V_w) / V_f]$$

Sulfate Concentration in Air-Dry Soil, (C): 144 ppm

Test Methods: OHD L49

Orig: HDR Engineering, Inc. (Kansas City, MO)
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Report On: Determining Soluble Sulfate Content in Soil

Lab No: 1967-1

Report No: 1967-1-1

File ID: 2230-0534

Acct ID: 0230HDR20

Page 1 of 1

Client: HDR Engineering, Inc.
Mr. Scott Goehri, P.E.
4435 Main Street, Suite 1000
Kansas City, MO 64111

Project: B2280 Chilled Water System & Water
Cooling Towers
OKC OK

Report Date: 10/27/2022

Location: B-2 (2.3'-3.8')

Sample Date: 10/26/2022

Sampled By:

Field ID: B-2

Test Method: OHD L-49

Material Represented: B-2 (2.3'-3.8')

Mass of Air-Dried Sample Passing #10 Sieve (W_s): 5.1 g

Mass of Deionized Water Used in Slurry (W_w): 200.12 g

Average Colorimeter Reading (R): 0.67 ppm

Volume of Original Filtrate Used (V_f): N/A ml

Volume of Deionized Water Added to Filtrate (V_w): N/A ml

Colorimeter Reading on Diluted Filtrate (R_d): N/A ppm

$$C = [R * (W_w / W_s)] \text{ or } [R_d * (W_w / W_s) * (V_f + V_w) / V_f]$$

Sulfate Concentration in Air-Dry Soil, (C): 26 ppm

Test Methods: OHD L49

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Report On: Determining Soluble Sulfate Content in Soil

Lab No: 1967-2

Report No: 1967-2-1

File ID: 2230-0534

Acct ID: 0230HDR20

Page 1 of 1

Client: HDR Engineering, Inc.
Mr. Scott Goehri, P.E.
4435 Main Street, Suite 1000
Kansas City, MO 64111

Project: B2280 Chilled Water System & Water
Cooling Towers
OKC OK

Report Date: 10/27/2022

Location: B-2 (8.8'-10.3')

Sample Date: 10/26/2022

Sampled By:

Field ID: B-2

Test Method: OHD L-49

Material Represented: B-2 (8.8'-10.3')

Mass of Air-Dried Sample Passing #10 Sieve (W_s): 5.06 g

Mass of Deionized Water Used in Slurry (W_w): 200.22 g

Average Colorimeter Reading (R): 0.67 ppm

Volume of Original Filtrate Used (V_f): N/A ml

Volume of Deionized Water Added to Filtrate (V_w): N/A ml

Colorimeter Reading on Diluted Filtrate (R_d): N/A ppm

$$C = [R * (W_w / W_s)] \text{ or } [R_d * (W_w / W_s) * (V_f + V_w) / V_f]$$

Sulfate Concentration in Air-Dry Soil, (C): 27 ppm

Test Methods: OHD L49

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